

Udayan Atreya

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EDUCATION

Master of Science, Computer Science

May 2026

San Jose State University, San Jose, US

Coursework: Applied Deep Learning, Retrieval Augmented Generation.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures & Algorithms, Database Management Systems, OOP.

TECHNICAL SKILLS

Web & APIs: FastAPI, REST APIs, .NET, AWS API Gateway, Git, CI/CD

Cloud & Databases: AWS (Lambda, S3, DynamoDB), MongoDB, Azure DevOps, DynamoDB

Languages: Python, Java, C#, SQL

AI/ML Frameworks: LangChain, RAG, HuggingFace, TensorFlow, PyTorch, RAGAS

LLMs & Tools: Claude, GPT, Gemini, Ollama, Llama, DeepSeek, Groq, Vapi

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Architected a user-facing web notification system by designing and integrating **RESTful APIs** (AWS API Gateway, Lambda) with a DynamoDB backend, enabling real-time frontend-to-backend data exchange and driving a **20% increase in daily active users** post-launch.
- Built a user-facing recommendation engine for **100+ automation assets**, surfacing personalized tool suggestions through clean, well-documented REST API endpoints, **reducing user discovery time by 30%**.
- Partnered with **cross-functional teams** to deliver a **web-based ChatOps platform**, integrating a natural-language frontend interface with backend automation workflows via RESTful microservices, **reducing overhead by 40%** (~5m to ~1m).
- Delivered a maintainable, well-documented code vulnerability remediation solution using clean serverless architecture (AWS Lambda, S3, EventBridge) and Azure DevOps CI/CD pipelines, safeguarding **400+ applications** across organizations.
- Optimized backend API performance by redesigning DynamoDB access patterns and CRUD service endpoints, **reducing data retrieval latency by 37%** to support high-performance, scalable web service delivery.
- Designed and documented a single-touch server patching solution via REST-driven serverless architecture (AWS, Azure DevOps), following clean-code best practices and **reducing MTTR by 80%**.

PROJECT EXPERIENCE

BMC Observability Platform (*FastAPI, LangChain, Gemini 1.5, MongoDB, AWS S3*)

[\[GitHub Link\]](#)

Jun 2025 – Jul 2025

- Developed a **web-based observability platform** featuring Grafana and Prometheus dashboards for real-time telemetry visualization, integrating a semantic router to achieve **95% query classification accuracy** and **reduce manual triage time by 30%**.
- Built **RESTful API endpoints** (FastAPI) connecting MongoDB and AWS S3 backends to a web frontend for real-time data exchange, accelerating root cause analysis by **cutting retrieval time by 75%** (2m to 30s).
- Implemented clean, well-documented policy enforcement for hardware interventions with role-based access control and human-in-the-loop validation, **preventing 100% of unauthorized state changes**.

Evaluator-Optimizer RAG (*LangChain, GPT-4o, Gemini 3 Pro, Llama-3, ChromaDB, Whisper*)

[\[GitHub Link\]](#)

Aug 2025 – Mar 2026

- Architected a modular, well-documented agentic RAG pipeline implementing an Evaluator-Optimizer feedback loop, **reducing hallucination by 25%** and **improving answer accuracy by 18%** over baselines for Language-Based Video Reasoning systems.
- Engineered an **RAG evaluation suite** using RAGAS and Llama-3-70b (LLM-as-a-judge) over 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Context Precision, Context Recall, and Semantic Similarity.
- Optimized the latency-accuracy trade-off via Dynamic Chunking and MMR search, **reducing context redundancy by 40%** while maintaining **90%+ semantic coverage**.